

HW2.1

Climate Modeling

SCIE 001 Math

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This

1. Body, i.e. large content block for the frame.

(a) That

$$\int_a^b$$

When approaching a modelling problem, mathematicians often start from fundamental principles (sometimes called first principles). We can think of these principles as the

where P_{in} and P_{out} denote the power corresponding to, respectively, energy flowing in from the Sun and energy flowing out as the Earth's thermal radiation.

2. **Your task:** Derive the EBM given above and verify that the units on both sides of the

Left Side	Right Side
$= JK^{-1} \frac{K}{s}$	$= W - W$
$= \frac{J}{s}$	$= W$
$= W$	
$W = W$	
Left Side = Right Side	

Therefore, the units are consistent.

$$\lim_{T \rightarrow 247^-} \alpha(T) = \alpha(247) \quad \lim_{T \rightarrow 247^+} \alpha(T) = \alpha(247) \quad \lim_{T \rightarrow 282^-} \alpha(T) = \alpha(282) \quad \lim_{T \rightarrow 282^+} \alpha(T) = \alpha(282)$$

3.

Your task: Whi

A Guide to Finding Intersection Points of P_{in} and P_{out} in Python

Create a new Jupyter notebook using anaconda. Copy and paste the following lines of code into a new cell.